

Advanced (Habanero)

Q1. Solve the equation $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve the equation $|2x - 3| = 7$.

- (A) $x = 2$ or $x = 5$
- (B) $x = -2$ or $x = -5$
- (C) $x = 2$ or $x = -5$
- (D) $x = 5$ or $x = -2$
- (E) $x = 0$

Q3. Solve the equation $|x + 2| = 4$.

- (A) $x = 2$
- (B) $x = -2$
- (C) $x = 2$ or $x = -6$
- (D) $x = -2$ or $x = 6$
- (E) $x = 0$

Q4. Solve the equation $|3x - 1| = 11$.

- (A) $x = 2$ or $x = 4$
- (B) $x = -2$ or $x = -4$
- (C) $x = 2$ or $x = -4$
- (D) $x = 4$ or $x = -2$
- (E) $x = 0$

Q5. Solve the equation $|x - 5| = 3$.

- (A) $x = 2$ or $x = 8$
- (B) $x = -2$ or $x = -8$
- (C) $x = 2$ or $x = -8$
- (D) $x = 8$ or $x = -2$
- (E) $x = 0$

Q6. Solve the equation $|2x + 5| = 9$.

- (A) $x = 1$ or $x = -7$
- (B) $x = -1$ or $x = 7$
- (C) $x = 1$ or $x = 7$
- (D) $x = -1$ or $x = -7$
- (E) $x = 0$

Q7. Solve the equation $|x + 1| = 6$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -7$
- (D) $x = -5$ or $x = 7$
- (E) $x = 0$

Q8. Solve the equation $|3x - 2| = 10$.

- (A) $x = 2$ or $x = 4$
- (B) $x = -2$ or $x = -4$
- (C) $x = 2$ or $x = -4$
- (D) $x = 4$ or $x = -2$
- (E) $x = 0$

Open Ended Questions (FRQ)

Q9. Solve the equation $|x - 3| = |2x + 1|$ and check for extraneous solutions.

Q10. Solve the equation $|x + 2| = |x - 5|$ and check for extraneous solutions.

Chef's Tips

- Remind students to check for extraneous solutions.
- Encourage students to use the definition of absolute value to solve equations.
- Emphasize the importance of considering both positive and negative cases when solving absolute value equations.